Predicting Firm Fast Growth: Classification Analysis

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1. Business Problem & Target Definition

We predict which firms will achieve fast growth, defined as sales increasing by more than 25% from 2012 to 2013. This threshold is grounded in corporate finance theory: growth exceeding 25% substantially outpaces typical economic expansion and indicates successful capital deployment. Alternative measures considered include absolute sales growth (favors larger firms) and two-year growth (reduces sample due to attrition). The 25% threshold balances identifying meaningful growth with maintaining sufficient positive cases (25.2% of firms) for robust modeling.

2. Data & Methodology

Sample: 21,723 Hungarian SMEs (€1K-€10M revenue) from 2012 base year, predicting 2013 outcomes. Data source: cs_bisnode_panel.csv (OSF).

Features (8 → 15 after encoding): log sales, firm age (linear & quadratic), new firm indicator, foreign management (≥50% ownership), industry category, CEO gender, region.

Models: Logistic Regression (L2 penalty, 5-fold CV), Random Forest (100 trees), Gradient Boosting (100 estimators, depth=3, learning rate=0.1).

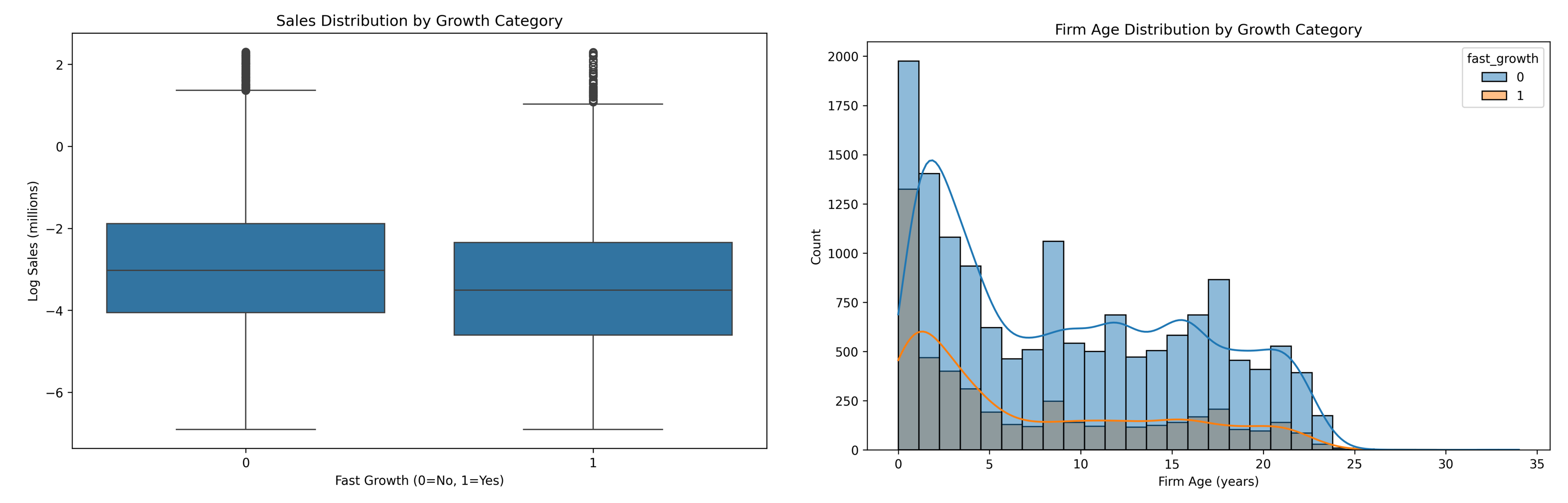

Figure 1: Sales and age distributions by growth category. Fast-growth firms show similar sales levels but younger age profile.

3. Probability Prediction Results

We evaluated three models using 5-fold cross-validation, measuring Area Under the ROC Curve (AUC):

Model	CV AUC	Std Dev	Interpretation
Logistic Regression	0.636	±0.008	Baseline linear model
Random Forest	0.580	±0.008	Underperforms, possible overfitting to noise
Gradient Boosting	**0.646**	**±0.010**	**Best discriminative power**

Decision: Gradient Boosting selected for highest AUC and consistent performance across folds.

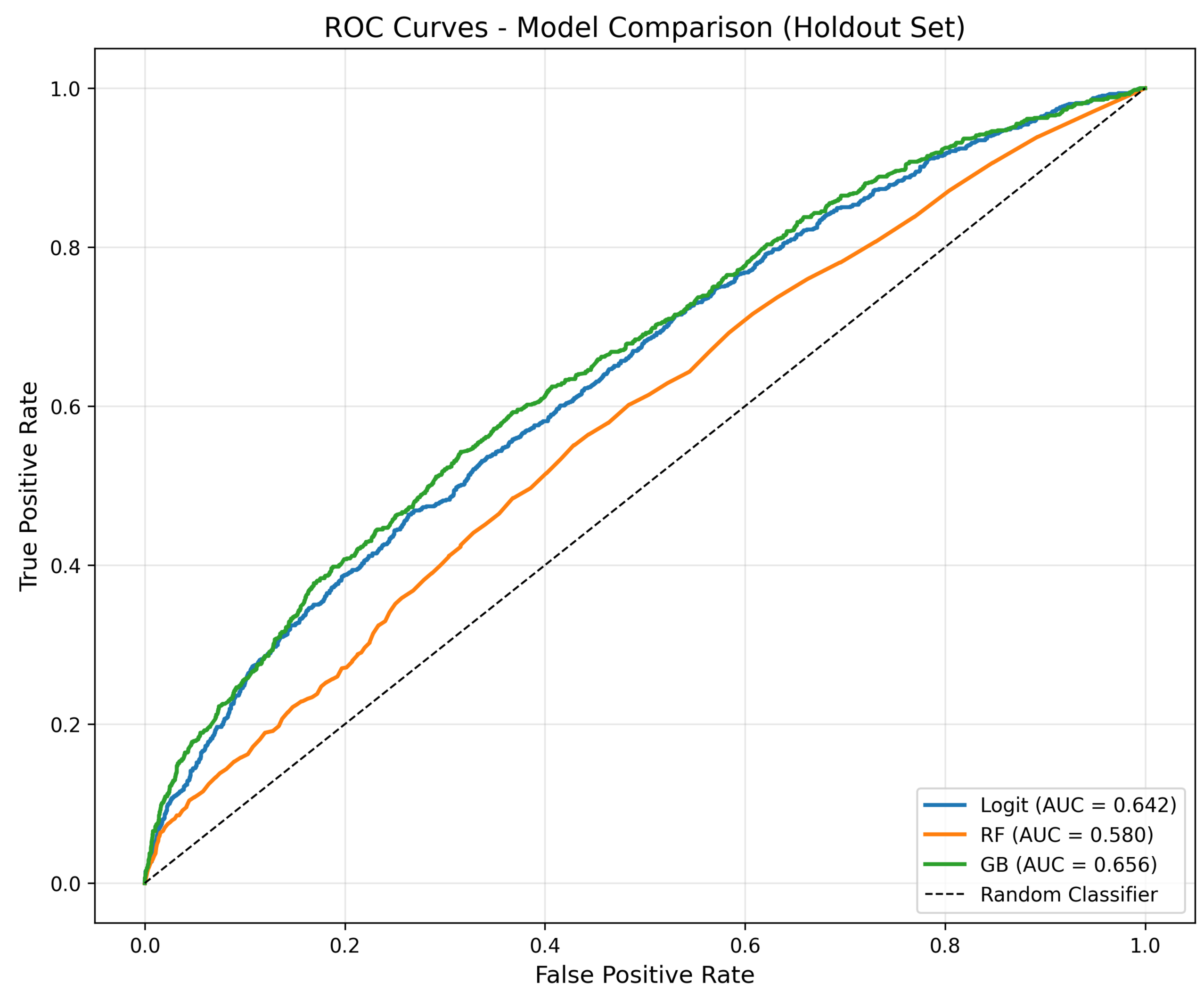

Figure 2: ROC curves on holdout set. Gradient Boosting (AUC=0.656) outperforms Logistic Regression (0.642) and Random Forest (0.580).

4. Classification with Loss Function

We defined business costs reflecting venture capital screening context: False Positive (investigating non-growth firm) = €1, False Negative (missing growth opportunity) = €5. This 5:1 ratio reflects that foregone investment returns substantially exceed due diligence expenses.

Model	Expected Loss	Optimal Threshold	vs. Best
Gradient Boosting	**2,894.60**	**0.14**	—
Logistic Regression	2,922.20	0.16	+1.0%
Random Forest	3,262.60	0.10	+12.7%

Result: Gradient Boosting minimizes expected loss by €27.60 vs. Logistic Regression and €368 vs. Random Forest.

5. Confusion Matrix & Model Performance

Evaluating the selected Gradient Boosting model on a holdout test set (20% stratified sample, threshold=0.18 optimized for loss):

	Predicted: No Growth	Predicted: Fast Growth
Actual: No Growth	911	2,060
Actual: Fast Growth	134	828

Performance Metrics:

- **Precision:** 28.7% (of predicted fast-growth, 28.7% are correct)
- **Recall:** 86.1% (model catches 86.1% of actual fast-growth firms)
- **Specificity:** 30.7% (correctly identifies 30.7% of no-growth firms)
- **Holdout Loss:** 2,730 (FP×1 + FN×5 = 2,060 + 670)

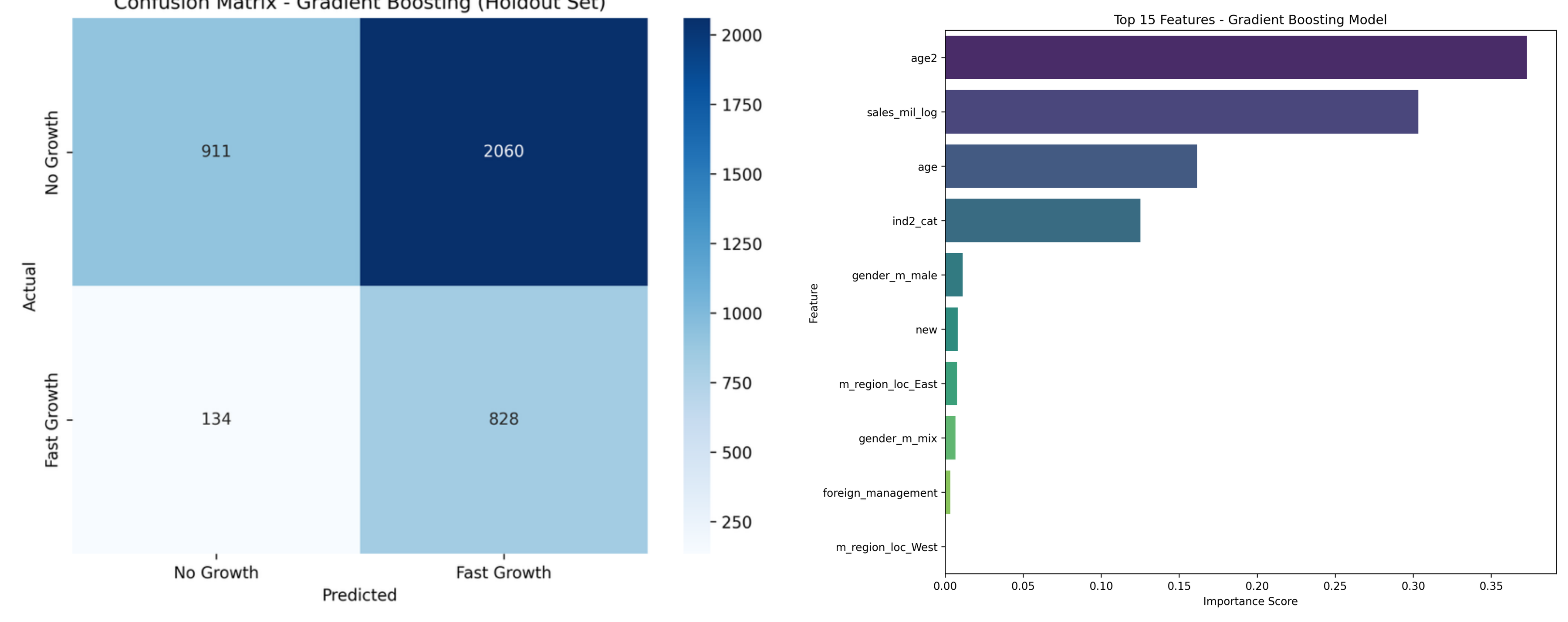

Figure 3: Confusion matrix heatmap and feature importance. Age² (37%) and sales magnitude (30%) dominate predictions.

Interpretation: The model is optimized for high recall—prioritizing catching growth opportunities over minimizing false alarms. This aligns with the business context where missing a fast-growth firm (FN) costs 5× more than investigating a false positive.

6. Industry-Specific Analysis

Applying the same loss function separately to manufacturing and services sectors:

Industry	Sample Size	Fast Growth %	Test AUC	Expected Loss
Manufacturing	5,752	27.1%	0.576	853
Services	13,721	23.3%	0.652	1,912

Key Finding: Services sector shows better predictability (AUC 0.652) compared to manufacturing (0.576). This likely reflects more standardized business models and customer interactions in services. Manufacturing exhibits higher growth prevalence (27.1% vs. 23.3%) but more variable growth patterns.

7. Recommendations

- **Investment Screening:** Use model to prioritize due diligence on firms scoring >0.18 probability. Expect 3.5:1 investigation ratio (investigate 3.5 firms to find 1 fast-grower).
- **Sector Focus:** Services firms offer more predictable growth patterns—allocate higher confidence to services sector predictions.
- **Age & Size Signals:** Younger firms (age²) and mid-sized companies (log sales) show strongest growth potential—target this profile actively.
- **Risk Tolerance:** Current threshold (0.18) minimizes loss under 5:1 cost ratio. Adjust threshold if opportunity costs change (lower threshold = higher recall, more FP).

8. Conclusion

The Gradient Boosting model achieves moderate predictive power (AUC 0.646) for identifying fast-growth firms, with performance optimized for high recall (86.1%) to minimize missed opportunities. The model is most useful as a screening tool to prioritize investigation efforts, rather than definitive classification. Firm age lifecycle effects (captured by quadratic age term at 37% importance) and current size (30% importance) are the dominant predictors. Industry-specific models reveal services sector offers more reliable growth prediction (AUC 0.652) than manufacturing (0.576). The analysis demonstrates that moderate-accuracy predictions, when aligned with business costs through loss function optimization, provide actionable value for investment decision support.

Data Source: cs_bisnode_panel.csv, Hungarian firm panel 2010-2015, accessed via OSF (https://osf.io/download/qsk8y/)
Code Repository: github.com/constantinosevag-rgb/Data-Analysis-3/tree/main/Assignment_2
Reproducibility: All analysis conducted with random_state=42 for full reproducibility.